

DUSSELDORF, Germany

WMO: 104000

Lat: 51.2958N

Lon: 6.7686E

Elev: 41

StdP: 100.84

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-4.1	-11.2	1.4	-4.2	-8.2	1.9	-2.5	12.9	9.3	11.4	9.1	3.3	50	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	31.1	19.7	28.9	18.9	26.9	18.2	20.9	27.8	20.1	26.3	19.2	25.0	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	23.2	18.0	13.0	22.1	17.1	12.2	21.4	60.5	28.4	57.3	26.5	54.6	25.0	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.1	8.1	-9.6	35.3	3.4	1.4	-12.0	36.3	-14.1	37.2	-16.0	38.0	-18.5	39.0		
			-10.1	22.8	3.3	1.0	-12.5	23.6	-14.5	24.2	-16.3	24.7	-18.8	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	3.2	3.7	6.8	11.1	14.2	17.7	19.9	18.7	15.9	12.1	7.6	4.8
	DBStd	6.85	4.19	4.00	4.07	3.82	3.89	3.36	3.51	3.05	3.08	3.70	3.66	4.00
	HDD10.0	816	211	176	112	32	8	0	0	0	0	21	91	166
	HDD18.3	2721	468	409	356	219	137	51	23	31	87	195	323	421
	CDD10.0	1311	2	1	14	64	138	232	307	269	177	86	18	3
	CDD18.3	172	0	0	0	1	9	33	72	42	14	2	0	0
	CDH23.3	1530	0	0	0	32	98	307	648	346	92	7	0	0
	CDH26.7	464	0	0	0	2	20	87	227	109	19	0	0	0
Wind	WSAvg	4.0	4.7	4.5	4.3	3.6	3.7	3.5	3.6	3.5	3.5	4.0	4.5	4.9
Precipitation	PrecAvg	863	75	66	58	53	64	78	83	88	67	68	79	84
	PrecMax	1112	162	142	113	111	138	180	147	191	164	200	133	175
	PrecMin	685	5	9	9	1	17	21	30	31	17	27	6	27
	PrecStd	96	36	34	31	25	31	36	36	45	38	32	32	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.1	20.1	26.9	29.2	32.2	35.0	33.1	29.8	24.1	17.8	14.2
		MCWB	10.5	8.1	11.7	14.9	18.7	20.4	20.5	20.5	19.0	17.2	14.3	11.4
	2%	DB	12.0	11.9	17.7	23.9	25.9	29.0	31.9	29.8	26.0	21.1	15.8	12.9
		MCWB	9.9	8.5	10.6	14.2	16.6	18.9	19.8	20.0	18.2	15.9	12.9	10.8
	5%	DB	10.2	10.2	15.1	21.0	23.9	26.9	29.1	26.9	23.8	19.0	14.1	11.2
		MCWB	8.6	8.0	10.1	13.2	16.0	18.2	18.8	18.5	17.6	14.9	11.8	9.2
	10%	DB	9.0	9.1	13.1	18.2	21.8	24.8	26.9	24.9	21.2	17.1	13.0	10.0
		MCWB	7.6	7.3	9.1	12.0	14.8	17.3	18.5	17.7	16.3	14.1	10.8	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	10.5	13.0	16.1	19.9	21.9	22.3	22.5	20.4	17.7	14.5	12.3
		MCDB	13.0	12.4	17.0	23.3	26.9	29.0	30.3	29.4	26.6	22.4	16.9	13.8
	2%	WB	10.1	9.2	11.7	14.7	18.3	20.3	20.9	20.6	19.1	16.3	13.1	11.0
		MCDB	11.4	10.9	15.9	21.7	23.6	26.3	27.9	27.4	24.6	20.0	15.2	12.7
	5%	WB	8.8	8.4	10.5	13.6	16.7	19.2	20.1	19.5	17.9	15.4	12.2	9.8
		MCDB	10.3	10.2	14.1	19.8	21.7	24.7	26.5	25.2	22.3	18.4	14.2	11.2
	10%	WB	7.4	7.1	9.5	12.6	15.4	18.2	19.3	18.5	16.9	14.4	11.0	8.6
		MCDB	8.6	8.8	12.7	17.8	20.5	23.3	25.4	23.3	20.7	17.1	13.0	9.8
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.5	8.0	10.1	9.7	9.8	9.8	9.3	8.8	7.1	5.3	4.4
		MCDBR	5.3	7.0	11.6	14.8	13.2	13.9	13.7	13.3	11.7	9.1	5.8	5.0
		MCWBR	4.7	4.8	6.6	7.1	6.2	6.1	4.7	5.6	5.6	5.1	4.3	4.1
	5% WB	MCDBR	5.1	6.3	9.7	12.8	11.1	11.5	11.5	11.0	10.2	7.9	5.1	4.8
		MCWBR	4.8	4.7	5.9	6.5	6.0	5.6	4.5	5.4	5.3	4.9	4.2	4.2
Clear-Sky Solar Irradiance	taub	0.326	0.349	0.378	0.414	0.419	0.419	0.428	0.420	0.392	0.364	0.342	0.321	
	taud	2.409	2.368	2.295	2.225	2.248	2.263	2.242	2.290	2.356	2.425	2.410	2.383	
	Ebn at Noon	689	770	812	823	837	838	823	809	790	743	659	627	
	Edh at Noon	61	82	106	127	130	129	130	118	99	77	61	54	
All-Sky Solar Radiation	RadAvg	0.74	1.41	2.53	4.07	4.90	5.29	5.03	4.29	3.14	1.85	0.87	0.56	
	RadStd	0.11	0.27	0.39	0.46	0.45	0.47	0.63	0.40	0.38	0.17	0.11	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page